Welcome: investigate, explain, then act

Understand / do

09:45-17:30; Azure beginners, experienced developers

Cost Management and observability are today's priorities

Fictional data; approved scopes; no shared fault injection

Verify / boundaries

Explain -> demonstrate -> practise -> verify

Record live / prepared query / offline review

Separate observations, hypotheses and verified fixes

Day 2 order / handouts: download the whole pack and keep its folder structure

Today's route and protected practice time

Understand / do

09:45 recap | 10:00 cost lab (75 min)

11:15 break | 11:30 Application Insights (75 min)

12:45 lunch | 13:30 KQL (75 min)

Verify / boundaries

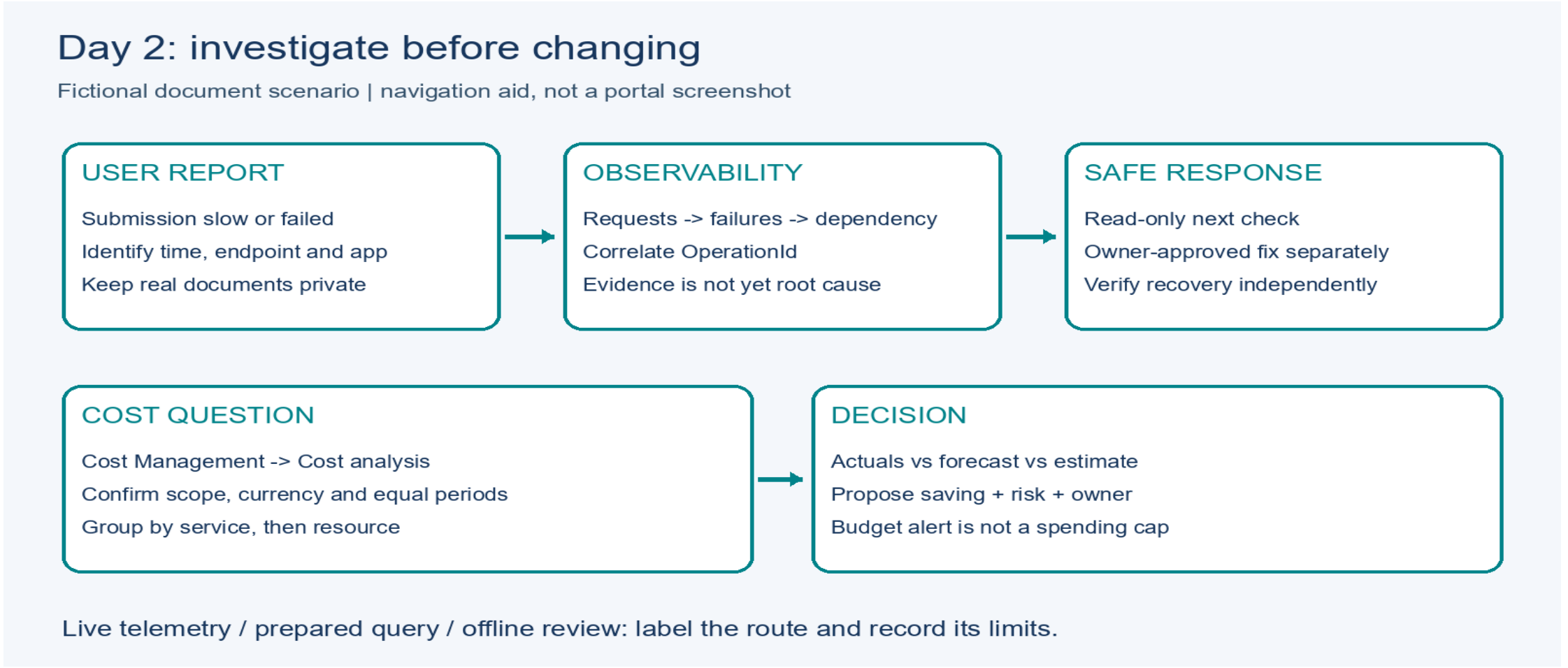
14:45 break | 15:00 Service Bus (35 min)

15:35 local tracker (45 min) | 16:20 team exercise (40 min)

17:00 assessment and cleanup | 17:30 close

Day 2 order / handouts: download the whole pack and keep its folder structure

One fictional story, three independent evidence routes



Day 2 order / handouts: download the whole pack and keep its folder structure

Cost: actual, amortised, forecast and estimate

Understand / do

Actuals report available billing data, which can lag

Amortised costs allocate commitments over time

Forecast predicts; calculator estimates a proposed design

Verify / boundaries

Record scope, currency, dates and metric

Compare equal complete periods with consistent filters

A blank cost view is not proof of no charges

Day 2 order / handouts: download the whole pack and keep its folder structure

Demo: Cost Management -> Scope -> Cost analysis

Understand / do

Confirm tenant and approved cost scope

Choose Services or a customisable view

Read dates, currency and cost metric before the chart

Verify / boundaries

Group by service, then resource

Compare equal periods; record filters

If denied, mark live access Blocked and use CSV

Day 2 order / handouts: download the whole pack and keep its folder structure

Practice: explain the largest cost increase

Understand / do

Open fictional-costs.csv if live data is unavailable

Sum each period separately; group by Service

Calculate absolute and percentage change

Verify / boundaries

Identify the main contributor and missing evidence

Propose one optimisation with risk and owner

Amounts are invented GBP values, not Azure prices

Day 2 order / handouts: download the whole pack and keep its folder structure

Mandatory budget design; optional real creation

Understand / do

Specify scope, currency, period, amount and owner

Choose approved recipients and notification thresholds

Preview in Budgets only if access permits

Verify / boundaries

No real creation without approval

Budget notifications do not automatically stop spend

Design, created budget and fired notification are different evidence

Day 2 order / handouts: download the whole pack and keep its folder structure

Cost decisions must preserve the service

Understand / do

App Service plan can charge while app is stopped

SQL storage and monitoring may continue

Access tier savings depend on retrieval and retention

Verify / boundaries

Do not disable security or required monitoring

Owner reviews changes and dependencies

Keep a resource-by-resource cleanup proposal

Day 2 order / handouts: download the whole pack and keep its folder structure

Application Insights, workspace and application

Understand / do

Requests describe incoming operations

Dependencies describe outgoing calls when collected

Workspace stores/query scope; component identifies app telemetry

Verify / boundaries

A connection string alone does not prove instrumentation

Local tracker is not connected to Azure Monitor

Use a prepared instrumented app or labelled dataset

Day 2 order / handouts: download the whole pack and keep its folder structure

Demo: navigate to an operation

Understand / do

Confirm assigned component and linked workspace

Inspect Performance / Failures in a bounded window

Record UTC request time and actual endpoint

Verify / boundaries

Open one request and transaction details

Inspect duration, status and operation ID

No configuration changes or shared failure injection

Day 2 order / handouts: download the whole pack and keep its folder structure

Read a request without overclaiming

Understand / do

Business status is not HTTP result code

A successful request can still be slow

A health endpoint may not query the database

Verify / boundaries

Explain actual row, time, source and scope

Live Metrics and stored logs can update differently

Absence of failure rows is not proof of health

Day 2 order / handouts: download the whole pack and keep its folder structure

Missing telemetry: investigate the collection path

Understand / do

Check endpoint, time, component and workspace

Check approved instrumentation with the owner

Verify dependency call occurred and is supported

Verify / boundaries

Wait for ingestion; retry bounded reads

No duplicate SDK, broad role or forced load

Use prepared dataset; record live status honestly

Day 2 order / handouts: download the whole pack and keep its folder structure

Observability cost: useful signal, bounded volume

Understand / do

Ingestion, retention, alerts and exports can charge

Duplicate instrumentation can duplicate volume

Caps may interrupt visibility; not a complete spend guarantee

Verify / boundaries

Five harmless requests, not a load test

No documents, secrets or bodies in evidence

Retention/sampling changes require owner review

Day 2 order / handouts: download the whole pack and keep its folder structure

KQL mode and schema come before the query

Understand / do

Logs -> close hub -> Agent off -> KQL mode

Workspace: AppRequests / AppDependencies

Application Insights: requests / dependencies

Verify / boundaries

Use matching query directory

Live workspace files need component resource ID

Prepared files need no live app; set historical time range

Day 2 order / handouts: download the whole pack and keep its folder structure

Level 1: first request rows

Understand / do

Start with AppRequests | take 10 conceptually

Add approved app filter and bounded time range

Run complete 01-first-requests.kql

Verify / boundaries

Prepared: five rows; live counts vary

Read Name, Success, result and duration

Save query text locally if central saving is unavailable

Day 2 order / handouts: download the whole pack and keep its folder structure

Duration and correlation names differ by scope

Understand / do

Workspace: TimeGenerated, DurationMs, OperationId

Resource: timestamp, duration, operation_Id

Resource duration / 1ms gives numeric milliseconds

Verify / boundaries

Supplied resource files normalise column names

Do not mix schemas or assume all fields are strings

Stored row count can differ from traffic under sampling

Day 2 order / handouts: download the whole pack and keep its folder structure

Level 2: filter failed requests

Understand / do

Run complete 02-failed-requests.kql

Filter Success == false after normalisation

Read request, result code, duration and operation ID

Verify / boundaries

Prepared rows: POST /documents and GET /private-status

Live can return no matching rows

A business Needs review outcome is not automatically HTTP failure

Day 2 order / handouts: download the whole pack and keep its folder structure

Level 3: slow dependency and related request

Understand / do

Run 03-slow-dependency-correlation.kql

Teaching threshold: dependency duration > 1000 ms

leftouter join relates requests by OperationId

Verify / boundaries

Prepared: op-003 and op-004

Slow failure and slow success need separate reasoning

Correlation guides next check; it does not prove root cause

Day 2 order / handouts: download the whole pack and keep its folder structure

Prepared data is a learning route, not live proof

Understand / do

Five requests, two failures, three dependencies

Fixed 28 October 2026 10:00-10:04 UTC

Run each whole Prepared file separately

Verify / boundaries

No ingestion, persistent table or app instrumentation

Offline CSV predictions are not KQL execution

Always record which route you completed

Day 2 order / handouts: download the whole pack and keep its folder structure

Incident note: observed -> hypothesis -> next check

Understand / do

Record app/source, UTC window and operation ID

Write request/dependency durations and status

State evidence separately from suggested cause

Verify / boundaries

Next check is bounded and read-only

Approved fixes and recovery verification are separate

Protect diagnostic value while considering cost

Day 2 order / handouts: download the whole pack and keep its folder structure

Service Bus: hand work off without waiting

Understand / do

Namespace contains queues; queue carries work

Producer sends; consumer receives and settles

Use tiny references, not document bodies

Verify / boundaries

Send is not business completion

Delivery can repeat; design idempotent effects

DLQ needs owner-approved investigation and replay

Day 2 order / handouts: download the whole pack and keep its folder structure

Demo: Explorer send -> peek -> receive

Understand / do

Use an isolated assigned queue with no consumer

Send one fictional message using approved data access

Peek inspects without settlement

Verify / boundaries

PeekLock requires Complete; ReceiveAndDelete is destructive

Check the actual offered mode before use

No Function trigger or local tracker integration

Day 2 order / handouts: download the whole pack and keep its folder structure

Queue practice and safe cleanup

Understand / do

Inspect your own message and record ID

Settle only your authorised test message

Refresh counts; allow for update delay

Verify / boundaries

Do not purge or use master/shared access keys

Do not create/upgrade namespace for a failed permission

Tier/capacity and operations can charge

Day 2 order / handouts: download the whole pack and keep its folder structure

Local tracker: what is actually implemented

Understand / do

Development-only localhost website

In-memory records and background queue/worker

No Azure SQL, Blob, Service Bus or telemetry connection

Verify / boundaries

Completed means training fields validated

Needs review is an expected business result

No real documents; no public binding or deployment

Day 2 order / handouts: download the whole pack and keep its folder structure

Run checks, then start the local website

Understand / do

Use .NET 10 SDK and the actual project folder

Run Checks with the absolute SampleData directory

Expect 14 PASS lines and exit code 0

Verify / boundaries

Start at localhost:5193 in Development

Check healthz and readyz separately

These endpoints do not prove Azure connectivity

Day 2 order / handouts: download the whole pack and keep its folder structure

Practice four documents; explain each outcome

Understand / do

Valid application -> Completed

Missing type -> Needs review

Reference mismatch -> Needs review

Verify / boundaries

Unsupported type -> Needs review

Refresh Queued/Processing until terminal outcome

Stop own host; in-memory records are temporary

Day 2 order / handouts: download the whole pack and keep its folder structure

Capstone: a failed submission and a higher bill

Understand / do

Use fictional evidence or assigned authorised live mode

Identify failure and a slow dependency

Compare equal-period costs by service

Verify / boundaries

Do not assume the cost change caused the request failure

No fault injection or configuration changes

40 minutes: investigate, challenge, hand over

Day 2 order / handouts: download the whole pack and keep its folder structure

Handover: evidence, trade-off and owner

Understand / do

Observation, operation ID and time range

Cost totals/delta and largest contributor

One hypothesis and one alternative explanation

Verify / boundaries

Next safe check and named owner

Optimisation benefit plus security/availability risk

Recovery and savings need independent verification

Day 2 order / handouts: download the whole pack and keep its folder structure

Security and networking: keep the independent checks

Understand / do

Network route, authentication and authorisation differ

Management roles are not universal data permissions

Use assigned app, queue and cost scopes

Verify / boundaries

No Owner role as a troubleshooting shortcut

No keys or personal data in notes

Optional identity/network handouts remain references

Day 2 order / handouts: download the whole pack and keep its folder structure

IaC, CI/CD and integration: what comes next

Understand / do

Versioned infrastructure can make deployments repeatable

Pipelines can build/test and deploy approved changes

Managed identity still needs scoped permissions

Verify / boundaries

Awareness only: no extra unprepared lab today

Target design is not a tested cloud implementation

Local tracker needs security, persistence and integration before hosting

Day 2 order / handouts: download the whole pack and keep its folder structure

Before/after knowledge check

Understand / do

Can you explain estimate vs actual vs forecast?

Does a budget stop spending?

How do workspace/resource query schemas differ?

Verify / boundaries

Can a success be slow? Does correlation prove cause?

Does Peek remove a message?

Which results were live, prepared or offline?

Day 2 order / handouts: download the whole pack and keep its folder structure

Cleanup is resource-specific, not one big delete

Understand / do

Stop only your own local host

Settle only your disposable message

Remove approved temporary budget only with owner agreement

Verify / boundaries

Do not delete shared resource groups or retained logs

App plans/supporting resources can still charge

Record owner and next action for each exception

Day 2 order / handouts: download the whole pack and keep its folder structure

Exit record and useful next steps

Understand / do

Show one reproducible query and an explained result

Present one cost proposal with risk

Record Passed / Blocked / Not run for each route

Verify / boundaries

No score substitutes for practical access evidence

Keep useful non-secret query text

Prepare missing live integration separately

Day 2 order / handouts: download the whole pack and keep its folder structure

Thank you - evidence before action

Understand / do

We inspected cost, requests, dependencies and messages

We practised realistic outcomes without unsafe changes

We distinguished working components from target designs

Verify / boundaries

Use the revised Day 2 order and whole folder structure

Bring scoped questions and honest limitations

Finish at 17:30; leave shared services unchanged

Day 2 order / handouts: download the whole pack and keep its folder structure